

Null and alternative hypotheses



- 1
 - a H_0 : The man is innocent.
 - b H_1 : The man is guilty of stealing.
- 2 $H_0 : \mu = 4000, H_1 : \mu > 4000$
- 3 $H_0 : \mu = 500, H_1 : \mu \neq 500$
- 4 $H_0 : \mu = 33, H_1 : \mu < 33$
- 5 $H_0 : \mu = 59\%, H_1 : \mu > 59\%$
- 6
 - a $H_0 : \mu = 2.5$
 - b $H_1 : \mu \neq 2.5$
- 7
 - a $H_0 : \mu = \$37.50$
 - b $H_1 : \mu > \$37.50$
- 8
 - a $H_1 : \mu = 2000$
 - b $H_1 : \mu > 2000$
- 9 $H_0 : \mu = 200, H_1 : \mu < 200$
- 10 $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 > \mu_2$
- 11
 - a The mean time of the boys is equal to the mean time of the girls.
 - b $\mu_1 = \mu_2$
 - c The mean time of the boys is different from the mean time of the girls.
 - d $\mu_1 \neq \mu_2$
- 12
 - a The colours of the smarties are evenly distributed.
 - b The colours of the smarties are not evenly distributed.
- 13 The amount of exercise per week is independent of a teacher's subject area.